RAG Knowledge Base Document

Section 1: Introduction to RAG

Retrieval-Augmented Generation (RAG) combines information retrieval with large language models to produce more accurate and grounded responses. The key components include document parsing, text chunking, embedding generation, vector storage, and retrieval.

Section 2: Document Parsing

PDF documents are parsed using tools like PDFPlumber or PyMuPDF.
Each page is analyzed to extract text, tables, and bounding boxes.
The coordinates are stored as (x, y, width, height) in point units.
A4 page size is 595.27 x 841.89 points.

Section 3: Chunking Strategy

Documents are split into overlapping chunks of 500-1000 tokens.
Each chunk retains its source metadata including page number and bounding box coordinates for precise PDF highlighting.

Section 4: Embedding and Retrieval

Text chunks are converted to dense vector embeddings using models like SentenceTransformers or OpenAI embeddings. These vectors are stored in a vector database (e.g., Milvus, Pinecone, FAISS) for similarity search.

Section 5: PDF Highlighting

When a retrieved chunk is displayed, its bounding box coordinates are converted to relative positions (x/page_width, y/page_height) and used to render highlight overlays on the PDF viewer.

Section 6: Frontend Rendering

The frontend uses PDF.js to render PDF pages. Highlight overlays are positioning using the relative coordinates calculated from the chunk bounding boxes and the PDF page dimensions.

End of Document